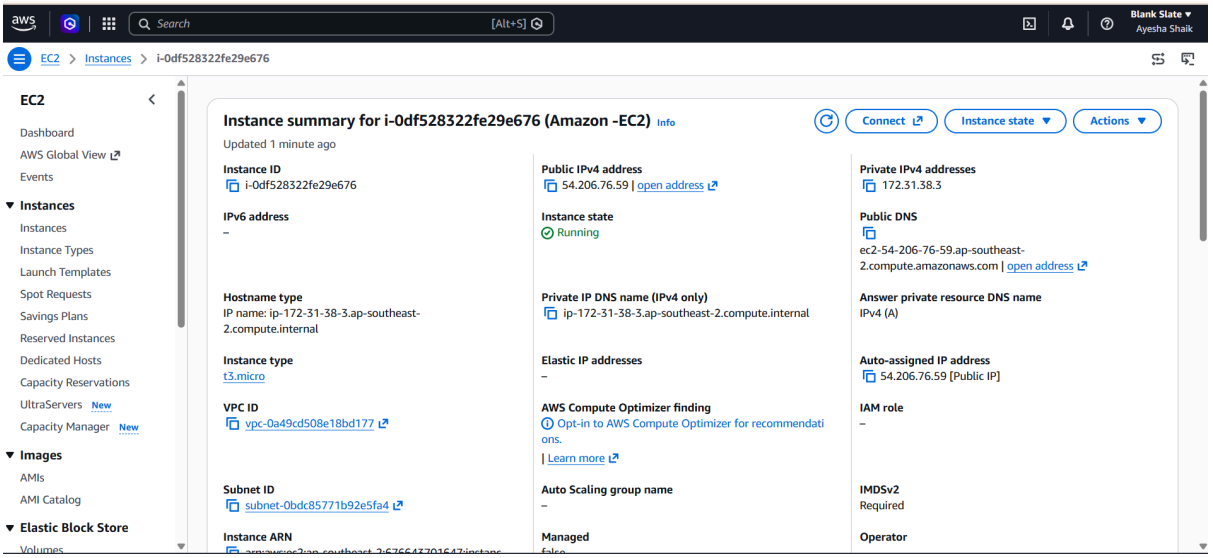
1. **Launch one EC2 instance using Amazon Linux 2.**



1. Log in to the **AWS Management Console**.
2. Open **EC2**.
3. Select the required **AWS Region**.
4. Click **Launch Instance**.
5. Enter the instance name
6. Under **Application and OS Images**, select **Amazon Linux 2**.
7. Select an appropriate instance type, such as `t2.micro`.
8. Select an existing **Key Pair** or create a new key pair.
9. Configure the **Security Group**:
    - SSH – `22`
    - HTTP – `80`
    - Custom TCP – `81`
    - Custom TCP – `8080`
    - Custom TCP – `8081`
10. Configure storage, for example **20 GiB**.
11. Review all the settings.
12. Click **Launch Instance**.
13. Go to **EC2 → Instances**.
14. Verify that the instance state is **Running**.
15. Verify that the status checks show **2/2 checks passed**.
16. Copy the **Public IPv4 address** for connecting to the EC2.

2. Install Docker

ec2-user@ip-172-31-38-3:~

```
SUPPORT_END="2029-06-30"
[ec2-user@ip-172-31-38-3 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository                   804 kB/s |  89 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-38-3 ~]$ sudo yum install docker
Last metadata expiration check: 0:00:18 ago on Tue Oct 27 09:08:04 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture  Version                    Repository          Size
================================================================================
Installing:
 docker                  x86_64        25.0.18-1.amzn2023.0.2     amazonlinux         49 M
Installing dependencies:
 container-selinux       noarch        4:2.245.0-1.amzn2023       amazonlinux         58 k
 containerd              x86_64        2.3.7-1.amzn2023.0.1       amazonlinux         24 M
 iptables-libs           x86_64        1.8.8-3.amzn2023.0.2       amazonlinux        401 k
 iptables-nft            x86_64        1.8.8-3.amzn2023.0.2       amazonlinux        183 k
 libcgroup               x86_64        3.0-1.amzn2023.0.1         amazonlinux         75 k
 libnetfilter_conntrack  x86_64        1.0.8-2.amzn2023.0.2       amazonlinux         58 k
 libnfnetlink            x86_64        1.0.1-19.amzn2023.0.2      amazonlinux         30 k
 libnftnl                x86_64        1.2.2-2.amzn2023.0.2       amazonlinux         84 k
 pigz                    x86_64        2.5-1.amzn2023.0.4         amazonlinux         83 k
 runc                    x86_64        1.4.3-1.amzn2023.0.3       amazonlinux        3.7 M

Transaction Summary
================================================================================
Install  11 Packages

Total download size: 77 M
Installed size: 290 M
Is this ok [y/N]: y
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm          1.8 MB/s |  58 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm             15 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm             5.3 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                  2.6 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm   2.1 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm            1.2 MB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                 2.8 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                       2.9 MB/s |  83 kB     00:00
(9/11): containerd-2.3.7-1.amzn2023.0.1.x86_64.rpm                69 MB/s |  24 MB     00:00
(10/11): runc-1.4.3-1.amzn2023.0.3.x86_64.rpm                     26 MB/s | 3.7 MB     00:00
(11/11): docker-25.0.18-1.amzn2023.0.2.x86_64.rpm                 69 MB/s |  49 MB     00:00
--------------------------------------------------------------------------------
Total                                                            102 MB/s |  77 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : runc-1.4.3-1.amzn2023.0.3.x86_64                      1/11
  Installing       : containerd-2.3.7-1.amzn2023.0.1.x86_64                2/11
  Running scriptlet: containerd-2.3.7-1.amzn2023.0.1.x86_64                2/11
  Installing       : pigz-2.5-1.amzn2023.0.4.x86_64                        3/11
  Installing       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                  4/11
```

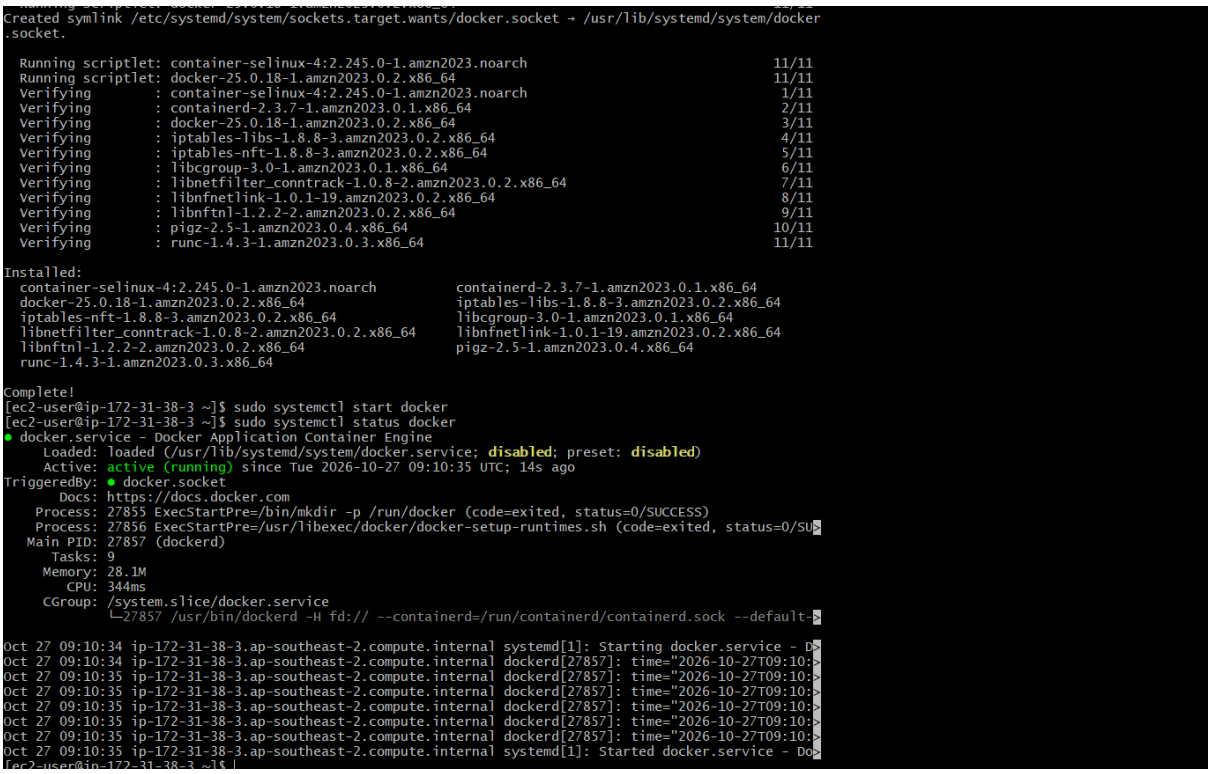
ec2-user@ip-172-31-38-3:~

```
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                 1.2 MB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                      2.8 MB/s |  84 kB     00:00
(8/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                            2.9 MB/s |  83 kB     00:00
(9/11): containerd-2.3.7-1.amzn2023.0.1.x86_64.rpm                     69 MB/s |  24 MB     00:00
(10/11): runc-1.4.3-1.amzn2023.0.3.x86_64.rpm                          26 MB/s | 3.7 MB     00:00
(11/11): docker-25.0.18-1.amzn2023.0.2.x86_64.rpm                      69 MB/s |  49 MB     00:00
--------------------------------------------------------------------------------------------------
Total                                                                 102 MB/s |  77 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Installing       : runc-1.4.3-1.amzn2023.0.3.x86_64                                      1/11
  Installing       : containerd-2.3.7-1.amzn2023.0.1.x86_64                                2/11
  Running scriptlet: containerd-2.3.7-1.amzn2023.0.1.x86_64                                2/11
  Installing       : pigz-2.5-1.amzn2023.0.4.x86_64                                        3/11
  Installing       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                  4/11
  Installing       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                             5/11
  Installing       : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                    6/11
  Installing       : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                             7/11
  Installing       : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                              8/11
  Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                              8/11
  Installing       : libcgroup-3.0-1.amzn2023.0.1.x86_64                                   9/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                        10/11
  Installing       : container-selinux-4:2.245.0-1.amzn2023.noarch                         10/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                         10/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                  11/11
  Installing       : docker-25.0.18-1.amzn2023.0.2.x86_64                                  11/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                  11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker
.socket.

  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                         11/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                  11/11
  Verifying        : container-selinux-4:2.245.0-1.amzn2023.noarch                          1/11
  Verifying        : containerd-2.3.7-1.amzn2023.0.1.x86_64                                 2/11
  Verifying        : docker-25.0.18-1.amzn2023.0.2.x86_64                                   3/11
  Verifying        : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                              4/11
  Verifying        : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                               5/11
  Verifying        : libcgroup-3.0-1.amzn2023.0.1.x86_64                                    6/11
  Verifying        : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                     7/11
  Verifying        : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                              8/11
  Verifying        : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                   9/11
  Verifying        : pigz-2.5-1.amzn2023.0.4.x86_64                                        10/11
  Verifying        : runc-1.4.3-1.amzn2023.0.3.x86_64                                      11/11

Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch          containerd-2.3.7-1.amzn2023.0.1.x86_64
  docker-25.0.18-1.amzn2023.0.2.x86_64                   iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
  iptables-nft-1.8.8-3.amzn2023.0.2.x86_64               libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64     libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
  libnftnl-1.2.2-2.amzn2023.0.2.x86_64                   pigz-2.5-1.amzn2023.0.4.x86_64
  runc-1.4.3-1.amzn2023.0.3.x86_64

Complete!
[ec2-user@ip-172-31-38-3 ~]$
```



1. Update the packages

sudo yum update -y

---

2. Install Docker

sudo amazon-linux-extras install docker -y

Start Docker:

sudo systemctl start docker

sudo systemctl enable docker

Check:

docker --version

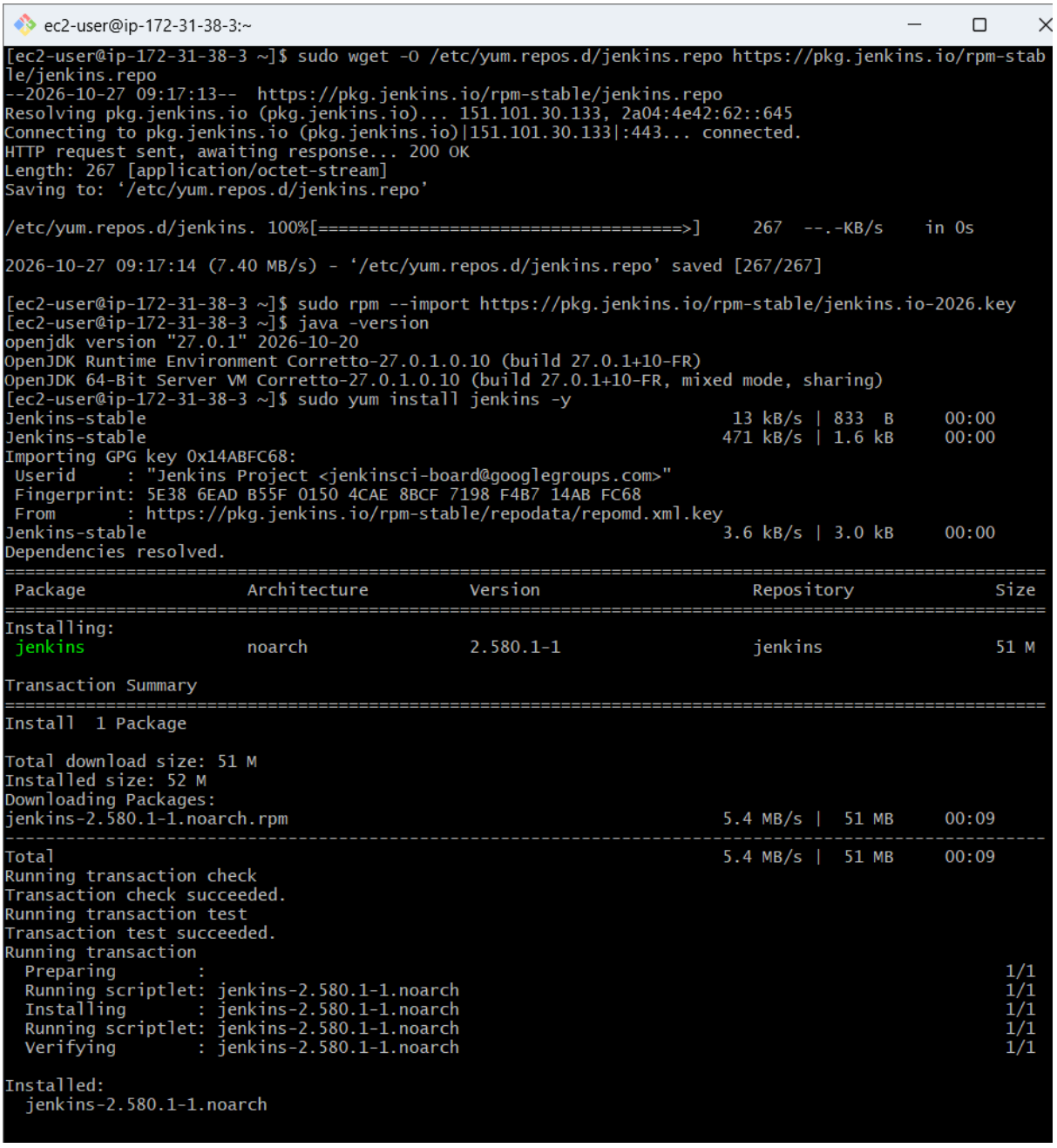
sudo systemctl status docker

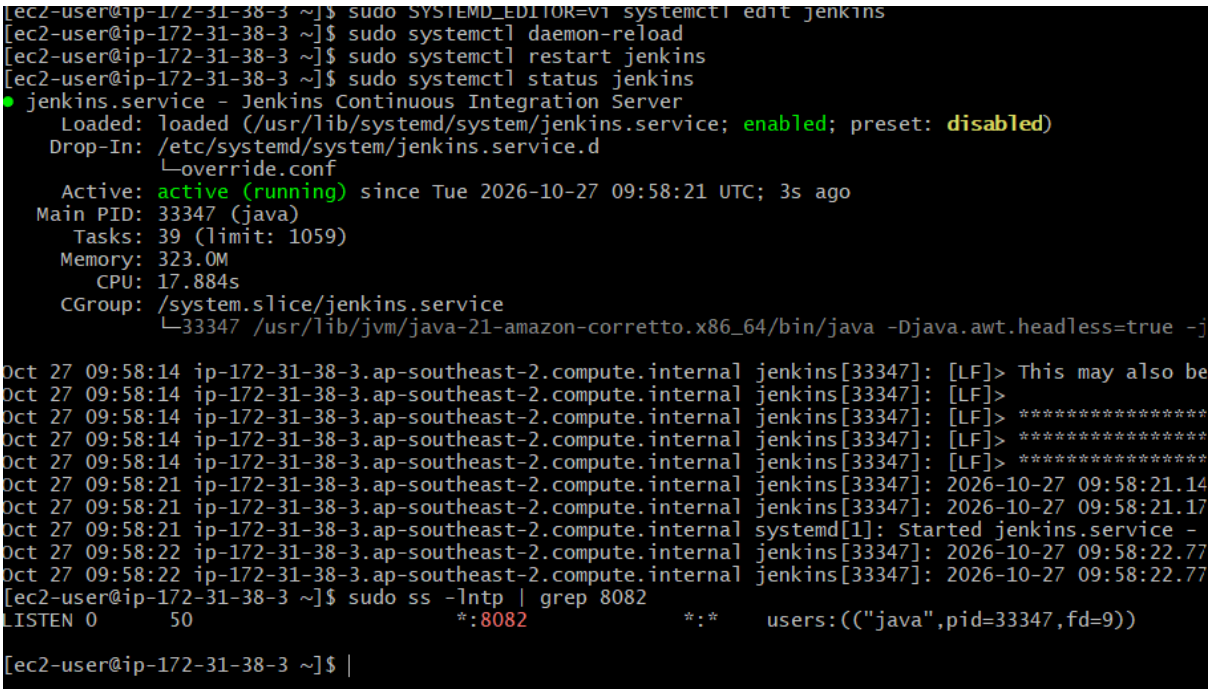
3. Install Jenkins

```
[ec2-user@ip-172-31-38-3 ~]$ sudo dnf install java
Last metadata expiration check: 0:05:19 ago on Tue Oct 27 09:08:04 2026.
Dependencies resolved.
================================================================================
 Package                          Arch       Version                      Repository       Size
================================================================================
Installing:
 java-27-amazon-corretto          x86_64     1:27.0.1.0.10-1.amzn2023.1   amazonlinux     217 k
Installing dependencies:
 alsa-lib                         x86_64     1.2.7.2-1.amzn2023.0.4       amazonlinux     503 k
 cairo                            x86_64     1.18.0-4.amzn2023.0.3        amazonlinux     717 k
 dejavu-sans-fonts                noarch     2.37-16.amzn2023.0.2         amazonlinux     1.3 M
 dejavu-sans-mono-fonts           noarch     2.37-16.amzn2023.0.2         amazonlinux     467 k
 dejavu-serif-fonts               noarch     2.37-16.amzn2023.0.2         amazonlinux     1.0 M
 fontconfig                       x86_64     2.13.94-2.amzn2023.0.2       amazonlinux     273 k
 fonts-filesystem                 noarch     1:2.0.5-12.amzn2023.0.2      amazonlinux     9.5 k
 freetype                         x86_64     2.13.2-5.amzn2023.0.2        amazonlinux     422 k
 giflib                           x86_64     5.2.1-9.amzn2023.0.4         amazonlinux      49 k
 google-noto-fonts-common         noarch     20240401-1.amzn2023.0.2      amazonlinux      17 k
 google-noto-sans-vf-fonts        noarch     20240401-1.amzn2023.0.2      amazonlinux     593 k
 graphite2                        x86_64     1.3.14-7.amzn2023.0.3        amazonlinux      97 k
 harfbuzz                         x86_64     7.0.0-2.amzn2023.0.2         amazonlinux     873 k
 java-27-amazon-corretto-headless x86_64     1:27.0.1.0.10-1.amzn2023.1   amazonlinux     110 M
 javapackages-filesystem          noarch     6.0.0-7.amzn2023.0.6         amazonlinux      12 k
 langpacks-core-font-en           noarch     3.0-21.amzn2023.0.4          amazonlinux      10 k
 libICE                           x86_64     1.1.1-3.amzn2023.0.1         amazonlinux      76 k
 libSM                            x86_64     1.2.4-3.amzn2023.0.1         amazonlinux      45 k
 libX11                           x86_64     1.8.10-2.amzn2023.0.2        amazonlinux     655 k
 libX11-common                    noarch     1.8.10-2.amzn2023.0.2        amazonlinux     144 k
 libXau                           x86_64     1.0.11-6.amzn2023.0.1        amazonlinux      33 k
 libXext                          x86_64     1.3.6-1.amzn2023.0.1         amazonlinux      42 k
 libXi                            x86_64     1.8.2-1.amzn2023.0.1         amazonlinux      42 k
 libXinerama                      x86_64     1.1.5-6.amzn2023.0.1         amazonlinux      16 k
 libXrandr                        x86_64     1.5.4-3.amzn2023.0.1         amazonlinux      29 k
 libXrender                       x86_64     0.9.11-6.amzn2023.0.2        amazonlinux      28 k
 libXt                            x86_64     1.3.0-3.amzn2023.0.1         amazonlinux     183 k
 libXtst                          x86_64     1.2.5-1.amzn2023.0.2         amazonlinux      21 k
 libbrotli                        x86_64     1.1.0-1.amzn2023.0.2         amazonlinux     333 k
 libjpeg-turbo                    x86_64     2.1.4-2.amzn2023.0.5         amazonlinux     190 k
 libpng                           x86_64     2:1.6.37-10.amzn2023.0.13    amazonlinux     128 k
 libxcb                           x86_64     1.17.0-1.amzn2023.0.1        amazonlinux     235 k
 pixman                           x86_64     0.43.4-1.amzn2023.0.4        amazonlinux     296 k
 xml-common                       noarch     0.6.3-56.amzn2023.0.2        amazonlinux      32 k

Transaction Summary
================================================================================
Install  35 Packages

Total download size: 119 M
Installed size: 324 M
Is this ok [y/N]: y
Downloading Packages:
```

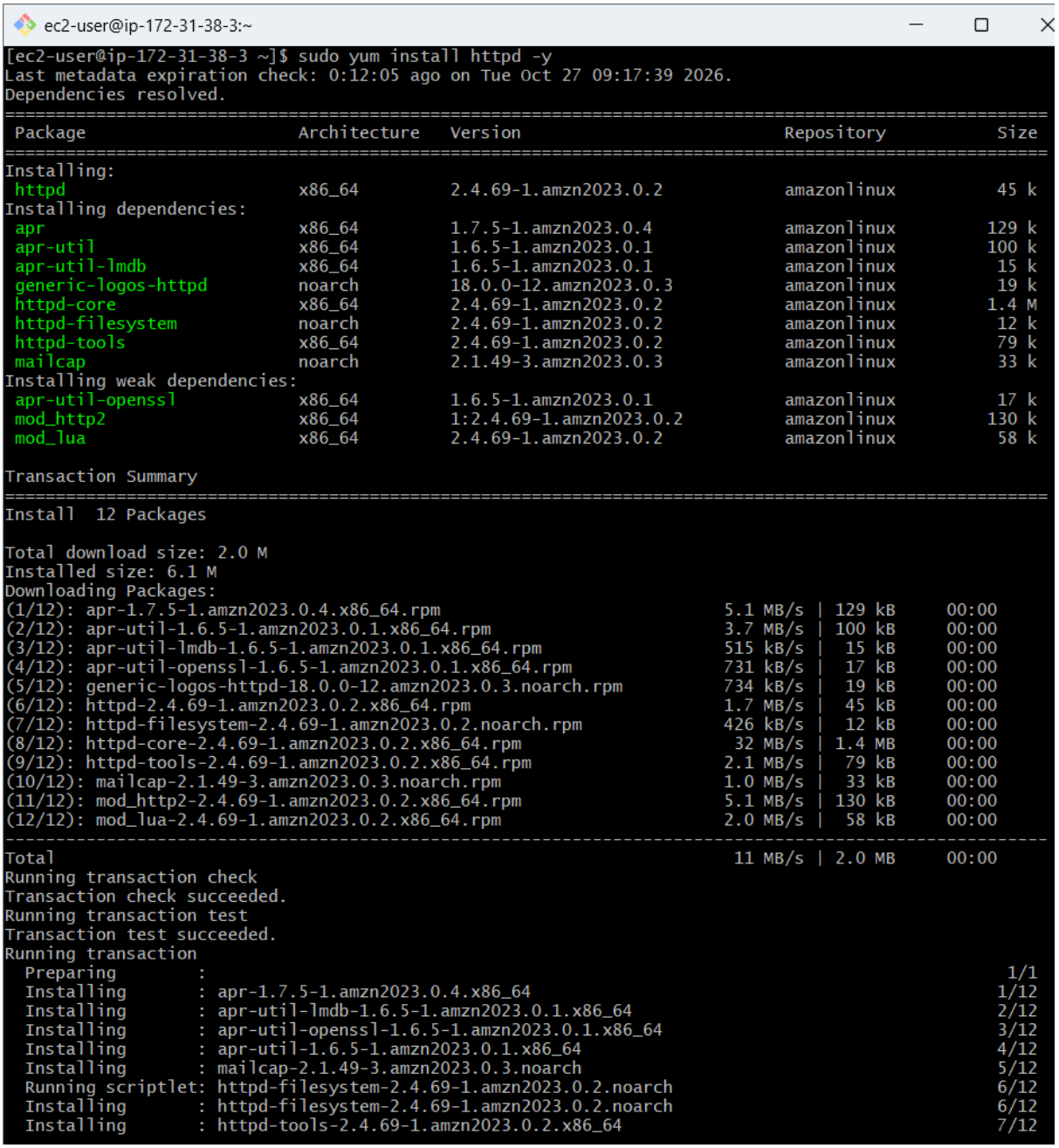

ec2-user@ip-172-31-38-3:~
[ec2-user@ip-172-31-38-3 ~]$ sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-10-27 09:17:13-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.30.133, 2a04:4e42:62::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.30.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'
/etc/yum.repos.d/jenkins. 100%[=====>] 267 --.-KB/s in 0s
2026-10-27 09:17:14 (7.40 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
[ec2-user@ip-172-31-38-3 ~]$ sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
[ec2-user@ip-172-31-38-3 ~]$ java -version
openjdk version "27.0.1" 2026-10-20
OpenJDK Runtime Environment Corretto-27.0.1.0.10 (build 27.0.1+10-FR)
OpenJDK 64-Bit Server VM Corretto-27.0.1.0.10 (build 27.0.1+10-FR, mixed mode, sharing)
[ec2-user@ip-172-31-38-3 ~]$ sudo yum install jenkins -y
Jenkins-stable 13 kB/s | 833 B 00:00
Jenkins-stable 471 kB/s | 1.6 kB 00:00
Importing GPG key 0x14ABFC68:
Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Jenkins-stable 3.6 kB/s | 3.0 kB 00:00
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
jenkins noarch 2.580.1-1 jenkins 51 M
Transaction Summary
Install 1 Package
Total download size: 51 M
Installed size: 52 M
Downloading Packages:
jenkins-2.580.1-1.noarch.rpm 5.4 MB/s | 51 MB 00:09
Total 5.4 MB/s | 51 MB 00:09
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Running scriptlet: jenkins-2.580.1-1.noarch 1/1
Installing : jenkins-2.580.1-1.noarch 1/1
Running scriptlet: jenkins-2.580.1-1.noarch 1/1
Verifying : jenkins-2.580.1-1.noarch 1/1
Installed:
jenkins-2.580.1-1.noarch



Jenkins Installation — Steps Only

1. Connect to the **Amazon Linux 2 EC2 instance**.
2. Update the system packages.
3. Install **Java 21**.
4. Set **Java 21 as the default Java version**.
5. Verify that Java 21 is active.
6. Add the **official Jenkins repository**.
7. Import the **Jenkins repository signing key**.
8. Install **Jenkins**.
9. Configure Jenkins to use **Java 21**.
10. Reload the system service configuration.
11. Enable the **Jenkins service** to start automatically after reboot.
12. Start the **Jenkins service**.
13. Verify that Jenkins is in the **Running** state.
14. Retrieve the **initial Jenkins administrator password**.
15. Allow **TCP port 8080** in the EC2 Security Group.
16. Open Jenkins using the EC2 public IP and port **8080**.
17. Enter the initial administrator password.
18. Complete the Jenkins initial setup.
19. Install the recommended Jenkins plugins.
20. Create the Jenkins administrator account.
21. Verify that the **Jenkins dashboard** is accessible.

4. Install Apache


ec2-user@ip-172-31-38-3:~
[ec2-user@ip-172-31-38-3 ~]$ sudo yum install httpd -y
Last metadata expiration check: 0:12:05 ago on Tue Oct 27 09:17:39 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
httpd x86_64 2.4.69-1.amzn2023.0.2 amazonlinux 45 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.5-1.amzn2023.0.1 amazonlinux 100 k
apr-util-lmdb x86_64 1.6.5-1.amzn2023.0.1 amazonlinux 15 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 19 k
httpd-core x86_64 2.4.69-1.amzn2023.0.2 amazonlinux 1.4 M
httpd-filesystem noarch 2.4.69-1.amzn2023.0.2 amazonlinux 12 k
httpd-tools x86_64 2.4.69-1.amzn2023.0.2 amazonlinux 79 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 33 k
Installing weak dependencies:
apr-util-openssl x86_64 1.6.5-1.amzn2023.0.1 amazonlinux 17 k
mod_http2 x86_64 1:2.4.69-1.amzn2023.0.2 amazonlinux 130 k
mod_lua x86_64 2.4.69-1.amzn2023.0.2 amazonlinux 58 k
Transaction Summary
Install 12 Packages
Total download size: 2.0 M
Installed size: 6.1 M
Downloading Packages:
(1/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 5.1 MB/s | 129 kB 00:00
(2/12): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm 3.7 MB/s | 100 kB 00:00
(3/12): apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64.rpm 515 kB/s | 15 kB 00:00
(4/12): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm 731 kB/s | 17 kB 00:00
(5/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 734 kB/s | 19 kB 00:00
(6/12): httpd-2.4.69-1.amzn2023.0.2.x86_64.rpm 1.7 MB/s | 45 kB 00:00
(7/12): httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch.rpm 426 kB/s | 12 kB 00:00
(8/12): httpd-core-2.4.69-1.amzn2023.0.2.x86_64.rpm 32 MB/s | 1.4 MB 00:00
(9/12): httpd-tools-2.4.69-1.amzn2023.0.2.x86_64.rpm 2.1 MB/s | 79 kB 00:00
(10/12): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm 1.0 MB/s | 33 kB 00:00
(11/12): mod_http2-2.4.69-1.amzn2023.0.2.x86_64.rpm 5.1 MB/s | 130 kB 00:00
(12/12): mod_lua-2.4.69-1.amzn2023.0.2.x86_64.rpm 2.0 MB/s | 58 kB 00:00
Total 11 MB/s | 2.0 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/12
Installing : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64 2/12
Installing : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64 3/12
Installing : apr-util-1.6.5-1.amzn2023.0.1.x86_64 4/12
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/12
Running scriptlet: httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch 6/12
Installing : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch 6/12
Installing : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64 7/12

```
  Installing       : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                6/12
  Installing       : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                     7/12
  Installing       : httpd-core-2.4.69-1.amzn2023.0.2.x86_64                      8/12
  Installing       : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                     9/12
  Installing       : mod_lua-2.4.69-1.amzn2023.0.2.x86_64                        10/12
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch           11/12
  Installing       : httpd-2.4.69-1.amzn2023.0.2.x86_64                          12/12
  Running scriptlet: httpd-2.4.69-1.amzn2023.0.2.x86_64                          12/12
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                              1/12
  Verifying        : apr-util-1.6.5-1.amzn2023.0.1.x86_64                         2/12
  Verifying        : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                    3/12
  Verifying        : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64                 4/12
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch            5/12
  Verifying        : httpd-2.4.69-1.amzn2023.0.2.x86_64                           6/12
  Verifying        : httpd-core-2.4.69-1.amzn2023.0.2.x86_64                      7/12
  Verifying        : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                8/12
  Verifying        : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                     9/12
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                        10/12
  Verifying        : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                    11/12
  Verifying        : mod_lua-2.4.69-1.amzn2023.0.2.x86_64                        12/12

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                   apr-util-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64         apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.69-1.amzn2023.0.2.x86_64
  httpd-core-2.4.69-1.amzn2023.0.2.x86_64           httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch
  httpd-tools-2.4.69-1.amzn2023.0.2.x86_64          mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64          mod_lua-2.4.69-1.amzn2023.0.2.x86_64

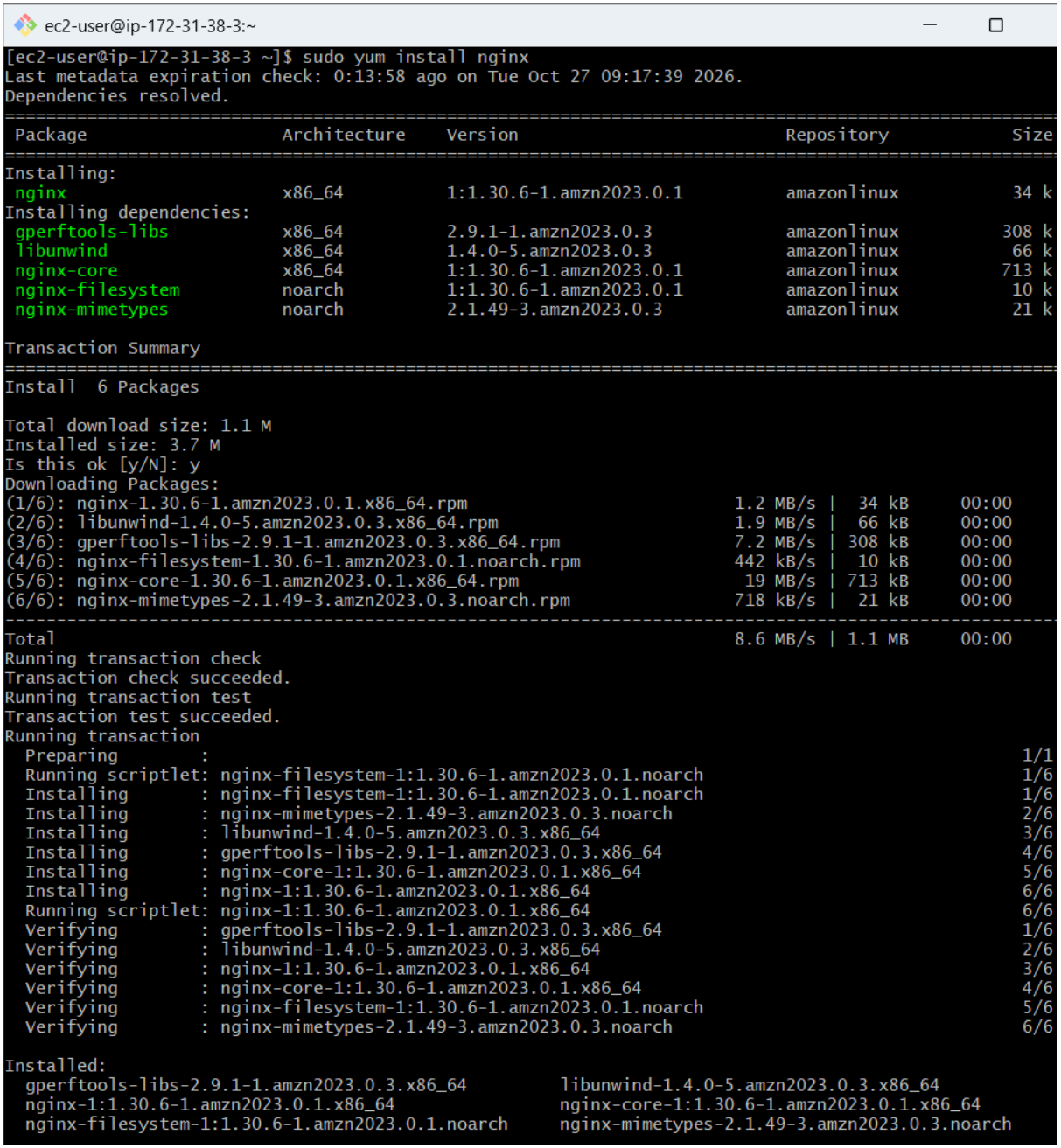
Complete!
[ec2-user@ip-172-31-38-3 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-38-3 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Tue 2026-10-27 09:29:53 UTC; 5s ago
       Docs: man:httpd.service(8)
   Main PID: 29945 (httpd)
     Status: "Started, listening on: port 80"
      Tasks: 177 (limit: 1059)
     Memory: 14.3M
        CPU: 68ms
     CGroup: /system.slice/httpd.service
             ├─29945 /usr/sbin/httpd -DFOREGROUND
             ├─29946 /usr/sbin/httpd -DFOREGROUND
             ├─29947 /usr/sbin/httpd -DFOREGROUND
             ├─29991 /usr/sbin/httpd -DFOREGROUND
             └─29992 /usr/sbin/httpd -DFOREGROUND

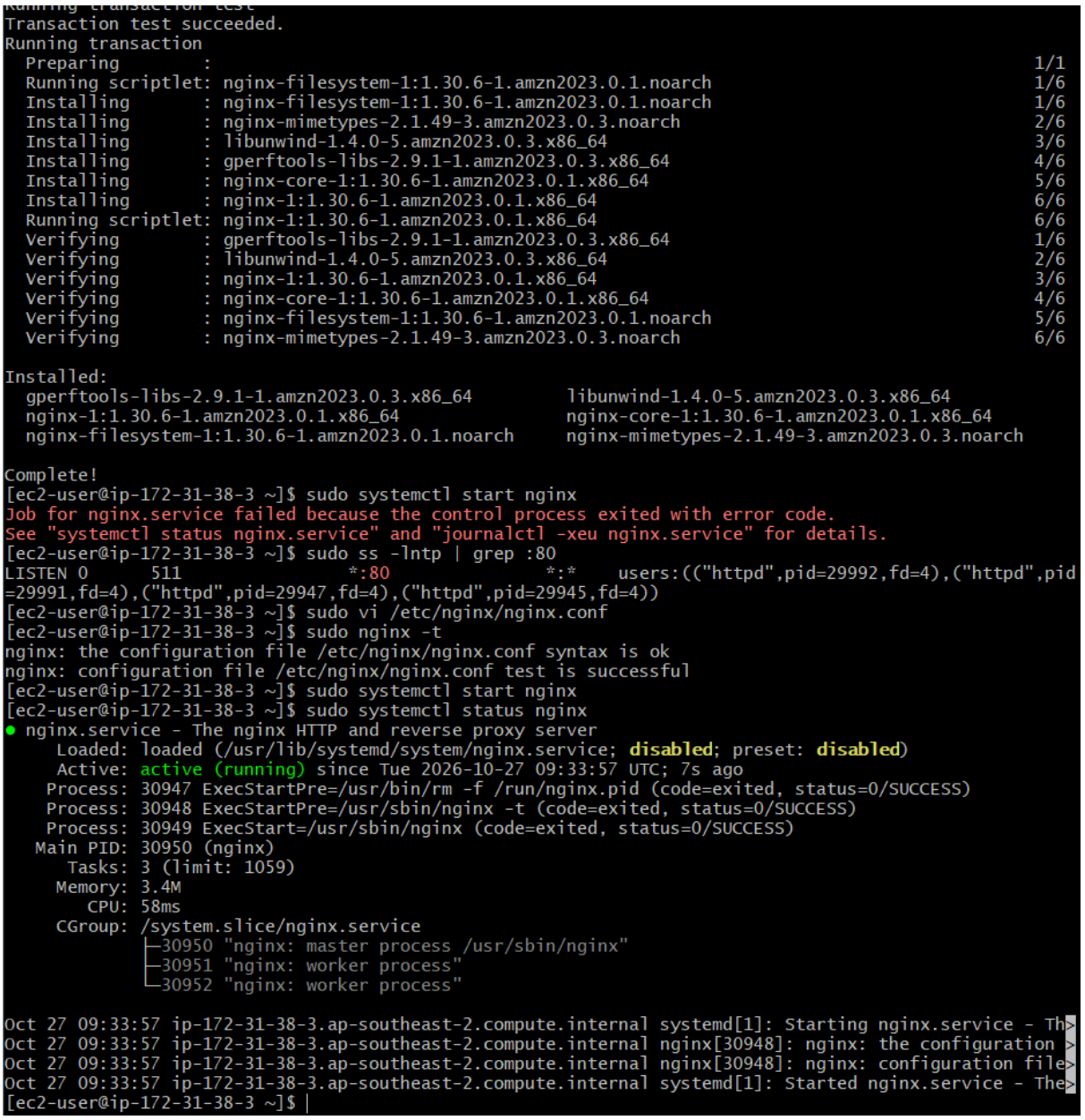
Oct 27 09:29:53 ip-172-31-38-3.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - Th>
Oct 27 09:29:53 ip-172-31-38-3.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - The>
Oct 27 09:29:53 ip-172-31-38-3.ap-southeast-2.compute.internal httpd[29945]: Server configured, listen>
lines 1-19/19 (END)
```

Install Apache on Amazon Linux 2 — Steps Only

1. Connect to the **Amazon Linux 2 EC2 instance**.
2. Update the system packages.
3. Install the **Apache HTTP Server** package (`httpd`).
4. Start the **Apache service**.
5. Enable Apache to start automatically after a reboot.
6. Verify that the Apache service is **running**.
7. Check that Apache is listening on **port 80**.
8. Allow **HTTP port 80** in the EC2 Security Group.
9. Open the EC2 **Public IPv4 address** in a browser.
10. Verify that the **Apache test page** is displayed.
11. Take a screenshot of the running Apache service and web page as evidence for the assignment.

5. Install Nginx





Nginx — Steps

1. Install Nginx:

sudo yum install nginx -y

2. Open Nginx configuration:

sudo vi /etc/nginx/nginx.conf

3. Change Nginx port from **80 to 81**.
4. Save and exit the file.
5. Test the Nginx configuration:

```
sudo nginx -t
```

6. Start Nginx:

```
sudo systemctl start nginx
```

7. Enable Nginx:

```
sudo systemctl enable nginx
```

8. Check Nginx status:

```
sudo systemctl status nginx
```

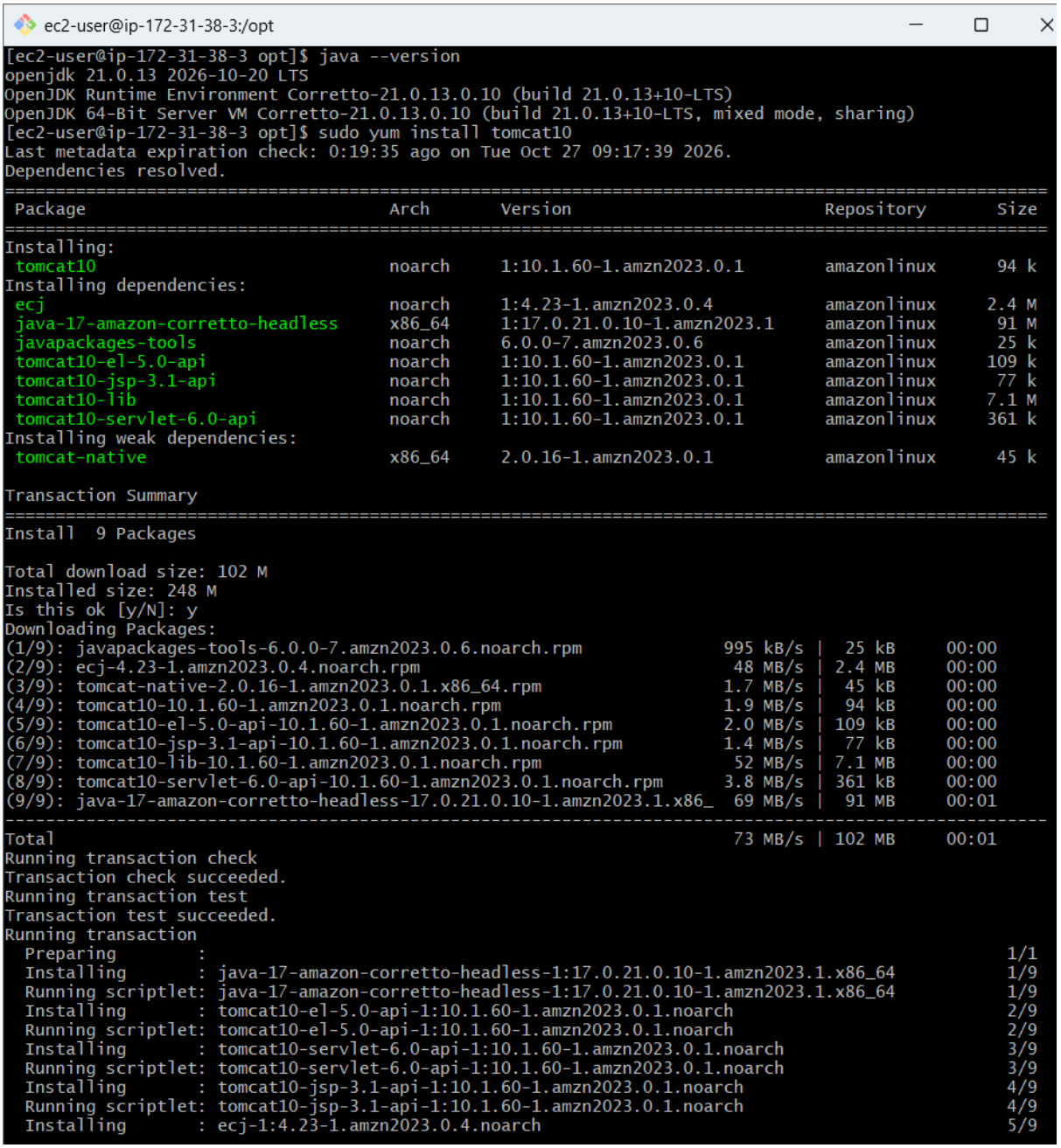
9. Verify Nginx is listening on port **81**:

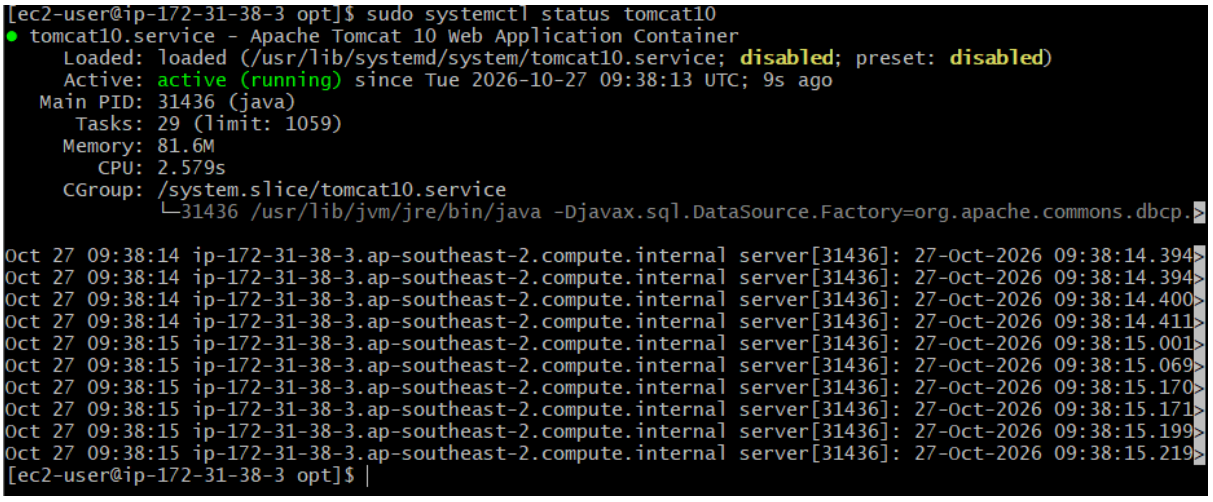
```
sudo ss -lntp | grep :81
```

10. Add **TCP port 81** to the EC2 Security Group.
11. Open in browser:

```
http://<EC2-Public-IP>:81
```

6. Install Apache Tomcat


ec2-user@ip-172-31-38-3:/opt
[ec2-user@ip-172-31-38-3 opt]$ java --version
openjdk 21.0.13 2026-10-20 LTS
OpenJDK Runtime Environment Corretto-21.0.13.0.10 (build 21.0.13+10-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.13.0.10 (build 21.0.13+10-LTS, mixed mode, sharing)
[ec2-user@ip-172-31-38-3 opt]$ sudo yum install tomcat10
Last metadata expiration check: 0:19:35 ago on Tue Oct 27 09:17:39 2026.
Dependencies resolved.
Package Arch Version Repository Size
Installing:
tomcat10 noarch 1:10.1.60-1.amzn2023.0.1 amazonlinux 94 k
Installing dependencies:
ecj noarch 1:4.23-1.amzn2023.0.4 amazonlinux 2.4 M
java-17-amazon-corretto-headless x86_64 1:17.0.21.0.10-1.amzn2023.1 amazonlinux 91 M
javapackages-tools noarch 6.0.0-7.amzn2023.0.6 amazonlinux 25 k
tomcat10-el-5.0-api noarch 1:10.1.60-1.amzn2023.0.1 amazonlinux 109 k
tomcat10-jsp-3.1-api noarch 1:10.1.60-1.amzn2023.0.1 amazonlinux 77 k
tomcat10-lib noarch 1:10.1.60-1.amzn2023.0.1 amazonlinux 7.1 M
tomcat10-servlet-6.0-api noarch 1:10.1.60-1.amzn2023.0.1 amazonlinux 361 k
Installing weak dependencies:
tomcat-native x86_64 2.0.16-1.amzn2023.0.1 amazonlinux 45 k
Transaction Summary
Install 9 Packages
Total download size: 102 M
Installed size: 248 M
Is this ok [y/N]: y
Downloading Packages:
(1/9): javapackages-tools-6.0.0-7.amzn2023.0.6.noarch.rpm 995 kB/s | 25 kB 00:00
(2/9): ecj-4.23-1.amzn2023.0.4.noarch.rpm 48 MB/s | 2.4 MB 00:00
(3/9): tomcat-native-2.0.16-1.amzn2023.0.1.x86_64.rpm 1.7 MB/s | 45 kB 00:00
(4/9): tomcat10-10.1.60-1.amzn2023.0.1.noarch.rpm 1.9 MB/s | 94 kB 00:00
(5/9): tomcat10-el-5.0-api-10.1.60-1.amzn2023.0.1.noarch.rpm 2.0 MB/s | 109 kB 00:00
(6/9): tomcat10-jsp-3.1-api-10.1.60-1.amzn2023.0.1.noarch.rpm 1.4 MB/s | 77 kB 00:00
(7/9): tomcat10-lib-10.1.60-1.amzn2023.0.1.noarch.rpm 52 MB/s | 7.1 MB 00:00
(8/9): tomcat10-servlet-6.0-api-10.1.60-1.amzn2023.0.1.noarch.rpm 3.8 MB/s | 361 kB 00:00
(9/9): java-17-amazon-corretto-headless-17.0.21.0.10-1.amzn2023.1.x86_ 69 MB/s | 91 MB 00:01
Total 73 MB/s | 102 MB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64 1/9
Running scriptlet: java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64 1/9
Installing : tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch 2/9
Running scriptlet: tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch 2/9
Installing : tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch 3/9
Running scriptlet: tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch 3/9
Installing : tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch 4/9
Running scriptlet: tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch 4/9
Installing : ecj-1:4.23-1.amzn2023.0.4.noarch 5/9


[ec2-user@ip-172-31-38-3 opt]$ sudo systemctl status tomcat10
tomcat10.service - Apache Tomcat 10 Web Application Container
Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; disabled; preset: disabled)
Active: active (running) since Tue 2026-10-27 09:38:13 UTC; 9s ago
Main PID: 31436 (java)
Tasks: 29 (limit: 1059)
Memory: 81.6M
CPU: 2.579s
CGroup: /system.slice/tomcat10.service
└─31436 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.
Oct 27 09:38:14 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:14.394
Oct 27 09:38:14 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:14.394
Oct 27 09:38:14 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:14.400
Oct 27 09:38:14 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:14.411
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.001
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.069
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.170
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.171
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.199
Oct 27 09:38:15 ip-172-31-38-3.ap-southeast-2.compute.internal server[31436]: 27-Oct-2026 09:38:15.219
[ec2-user@ip-172-31-38-3 opt]$

Tomcat 10 — Steps

1. Install Tomcat:

```
sudo yum install tomcat10 -y
```

2. Check the installed service name:

```
sudo systemctl status tomcat10
```

3. Start Tomcat:

```
sudo systemctl start tomcat10
```

4. Enable Tomcat:

```
sudo systemctl enable tomcat10
```

5. Check status:

```
sudo systemctl status tomcat10
```

6. Check the Tomcat port:

```
sudo ss -lntp | grep java
```

7. Because Jenkins uses **8080**, change Tomcat from **8080** → **8081** in its configuration.
8. Restart Tomcat:

```
sudo systemctl restart tomcat10
```

9. Verify port 8081:

```
sudo ss -lntp | grep :8081
```